MATH 214 - Assignment 1

Numerical Root-Finding Methods
Bisection, Fixed-Point Iteration, and Newton's Method
Spring 2026

1. Introduction

This report presents the implementation and comparative analysis of three classical root-finding algorithms: (1) Bisection Method, (2) Fixed-Point Iteration, and (3) Newton's Method.

All solvers use tolerance eps = 1e-6 and a maximum of 100 iterations. Stopping criteria: |x_{k+1} - x_k| < eps or |f(x_k)| < eps.

Source code: solvers.py (modular solver functions), main.py (driver & plots).

2. Results

2.1 Problem 1: f(x) = x^3 - x - 2

Unique real root: x* = 1.5213797068.

Method	Init	Iters	Root	|f(root)|
Bisection	[1, 2]	20	1.5213804245	4.27e-06
FP (good)	g=(x+2)^(1/3)	7	1.5213796793	1.64e-07
FP (bad)	g=x^3-2	7(div)	-8.41e+23	5.95e+71
Newton	x0=1.5	2	1.5213798060	5.89e-07

Observations:

- Newton converged in just 2 iterations (quadratic convergence).
- g(x)=(x+2)^(1/3) has |g'(x*)|~0.16<1, converging in 7 iterations.
- g(x)=x^3-2 has |g'(x*)|=3x*^2~6.94>>1, causing immediate divergence.
- Bisection converges reliably but slowly (20 iterations, linear).

2.2 Problem 2: f(x) = cos(x) - x

Unique root (Dottie number): x* = 0.7390851332.

Method	Init	Iters	Root	|f(root)|
Bisection	[0, 1]	20	0.7390851974	1.08e-07
Fixed-Point	g=cos(x),x0=0.5	33	0.7390855939	7.71e-07
Newton	x0=0.0	4	0.7390851334	2.85e-10
Newton	x0=1.0	3	0.7390851334	2.85e-10
Newton	x0=5.0	20	0.7390851447	1.93e-08
Newton	x0=10.0	92	0.7390851332	3.21e-13

Observations:

- Newton converges in 3-4 steps for close initial guesses (x0=0, 1).

- For x0=5 or 10, Newton needs many more iterations due to oscillatory cos(x).
- Fixed-point g=cos(x) converges (|g'|~0.67<1) but slowly at 33 iterations.

2.3 Problem 3: f(x) = x^3 (triple root at x=0)

Method	Init	Iters	Root	|f(root)|
Bisection	[-0.5, 1]	6	-0.0078125000	4.77e-07
Fixed-Point	g=-x^3+x,x0=0.5	100(fail)	0.0689415485	3.28e-04
Newton	x0=1.0	12	0.0077073466	4.58e-07

Convergence ratio analysis (Newton):

k	|e_k|	|e_k|/|e_{k-1}|
2	4.44e-01	0.6667
3	2.96e-01	0.6667
4	1.98e-01	0.6667
5	1.32e-01	0.6667
8	3.90e-02	0.6667
12	7.71e-03	0.6667

Observations:

- Bisection converges quickly (6 iters) since f(x)=x^3 has a sign change on [-0.5,1].
- Fixed-Point with g=-x^3+x failed (100 iters). |g'(0)|=1, violating contraction.
- Newton degrades to LINEAR convergence: ratio |e_k|/|e_{k-1}| = 2/3 = (m-1)/m for m=3.

3. Convergence Plots

All plots use semilogy scale: log|x_k - x*| vs. iteration count.

Plot 1 - Problem 1: f(x) = x^3 - x - 2

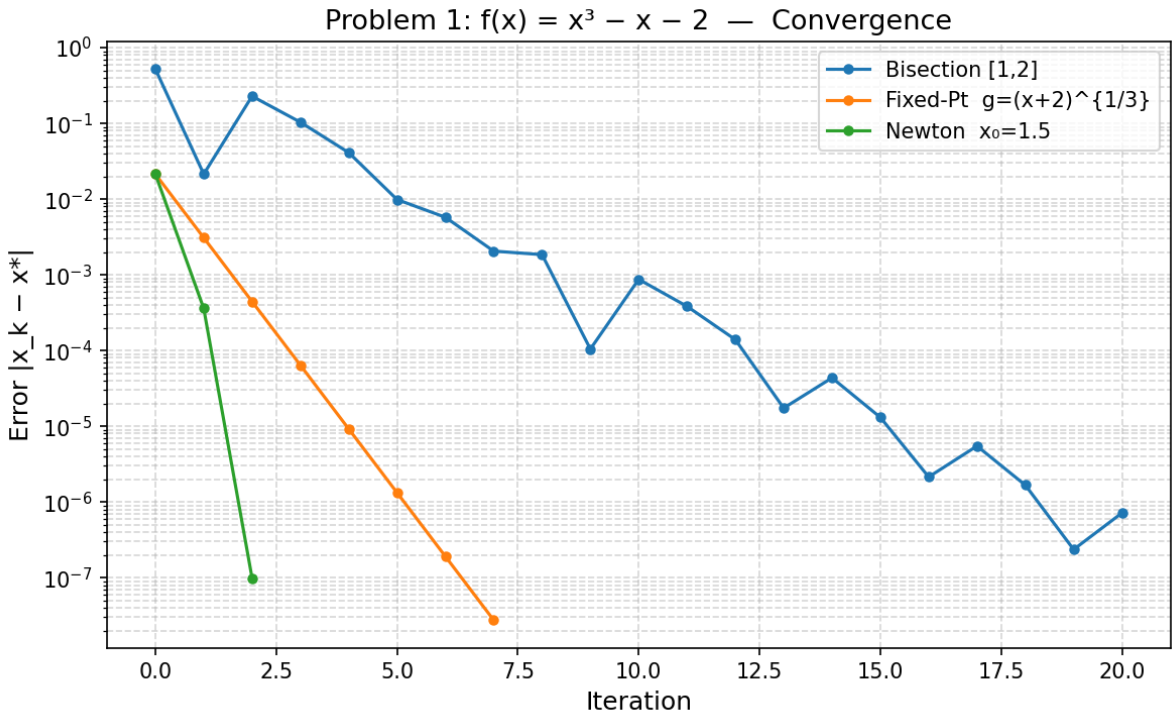

Figure 1: Problem 1 convergence comparison

Plot 2 - Problem 2: f(x) = cos(x) - x

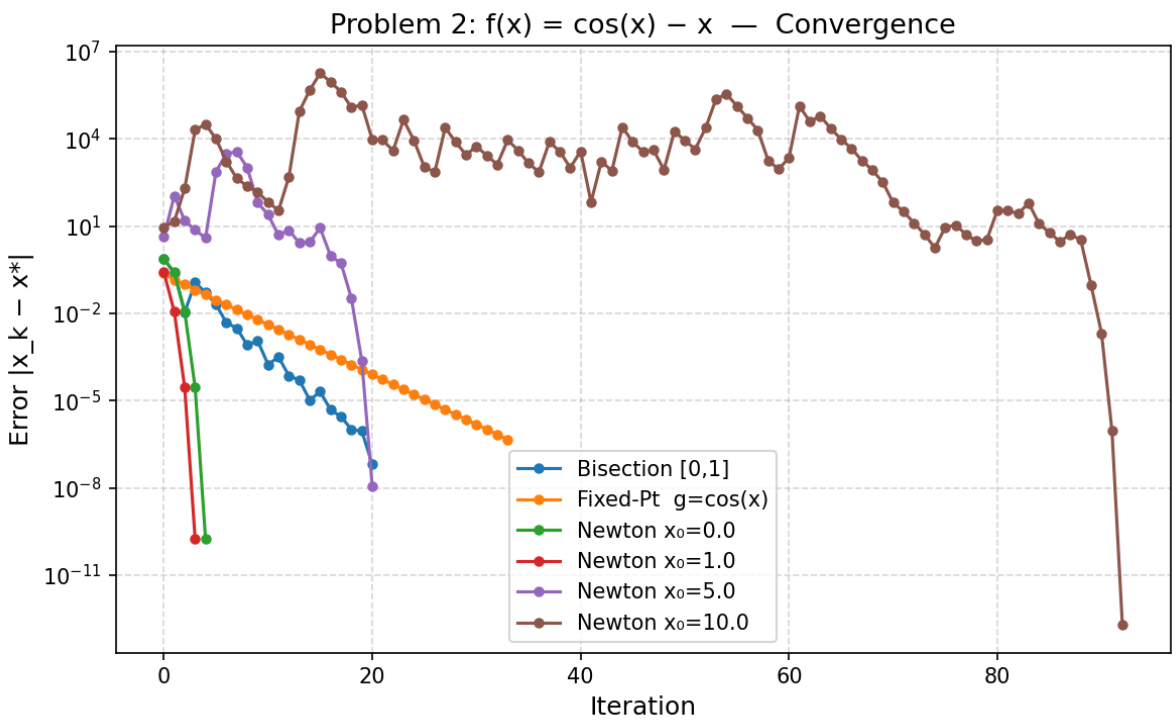

Figure 2: Problem 2 convergence with different Newton initial guesses

Plot 3 - Problem 3: f(x) = x^3 (all three methods)

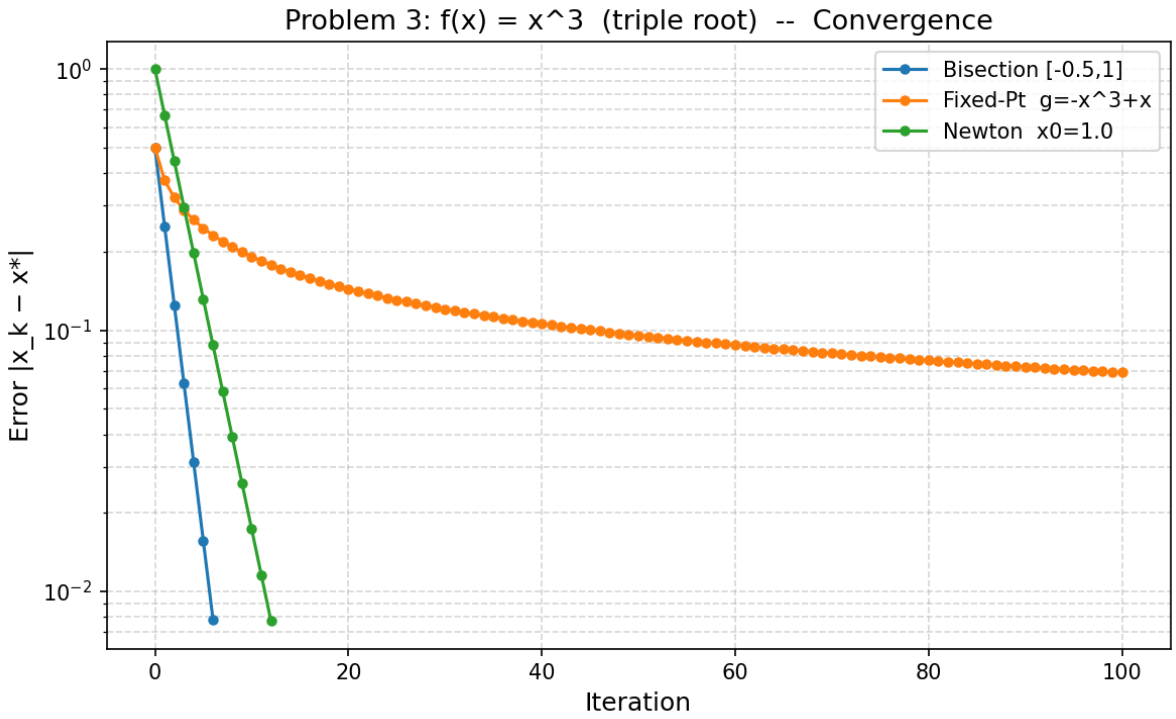

Figure 3: All three methods on triple root

Plot 4 - Divergent Fixed-Point (Problem 1)

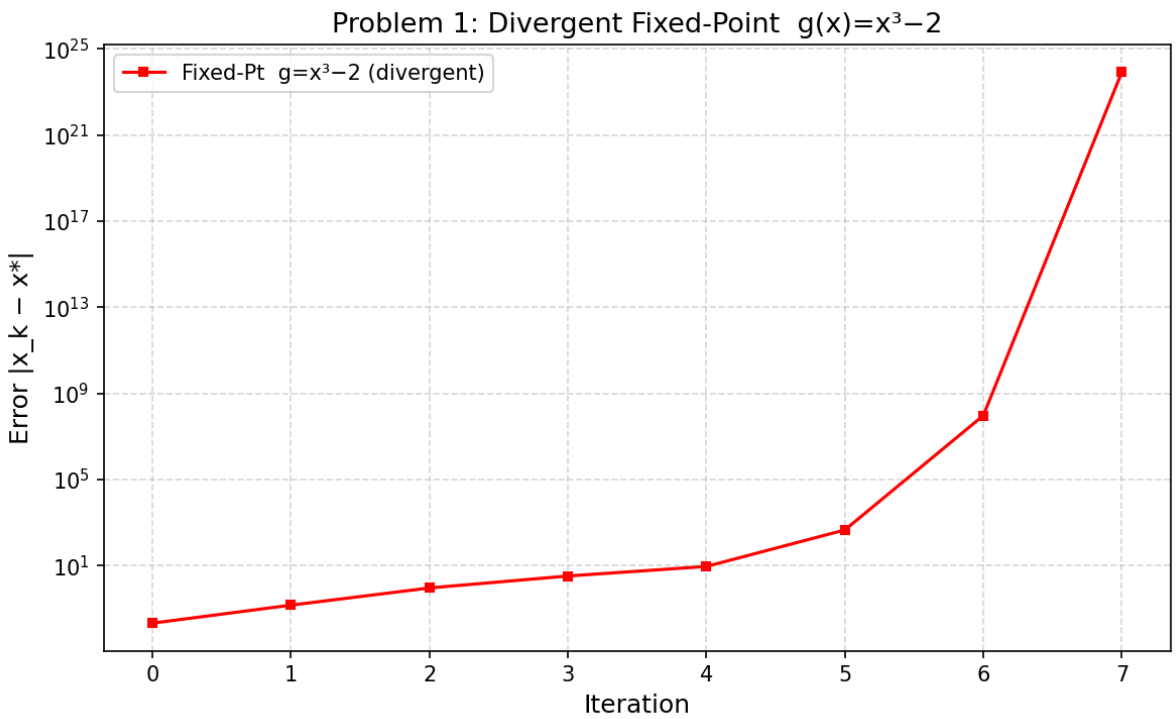

Figure 4: Divergence of g(x)=x^3-2

Plot 5 - Newton Ratio for x^3

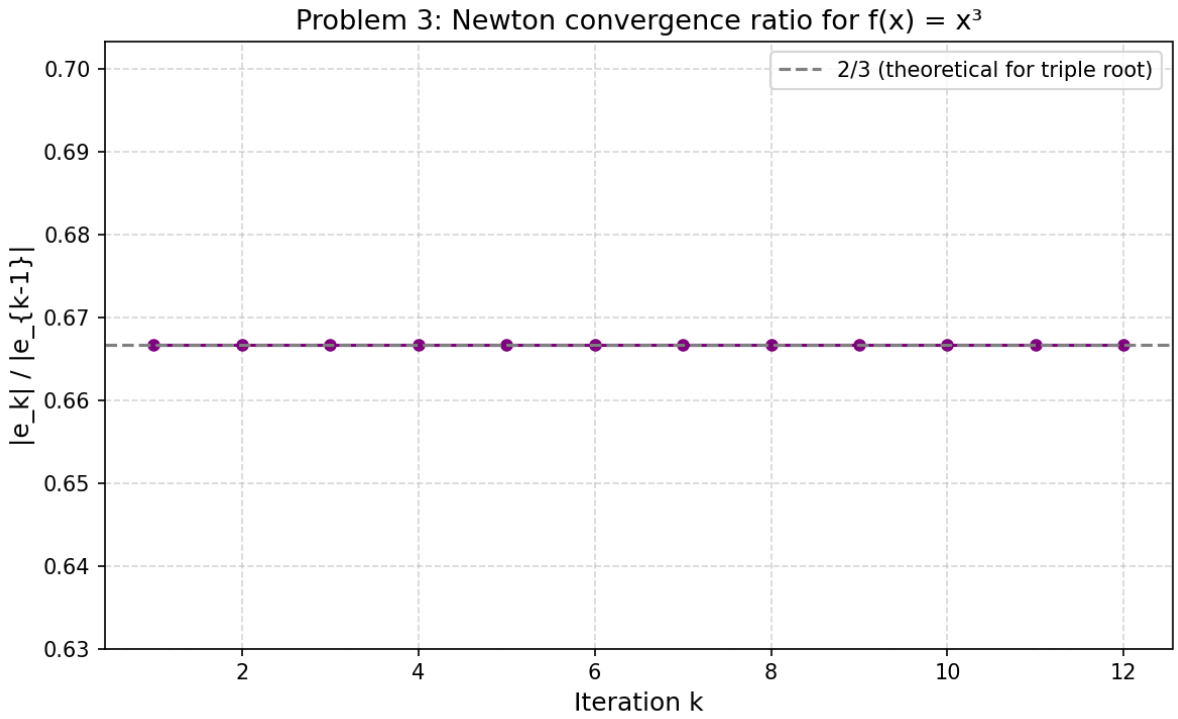

Figure 5: Constant ratio 2/3 confirms linear convergence

4. Discussion

4.1 Most robust and fastest method

Most robust: Bisection.

Given a valid bracket with f(a)*f(b)<0, bisection is guaranteed to converge. It cannot diverge or be affected by derivative issues.

Fastest: Newton's method.

For simple roots with good initial guesses, Newton achieves quadratic convergence, converging in 2-4 iterations. However, it can diverge or slow down when the derivative is small, the initial guess is far, or the root has multiplicity > 1.

4.2 Failure cases

Newton on f(x) = x^3:

The update x_{k+1} = x_k - x_k^3/(3x_k^2) = (2/3)*x_k shows that near the triple root x*=0, f'(x) = 3x^2 -> 0, causing the step to underestimate the correction. Convergence degrades from quadratic to linear with ratio (m-1)/m = 2/3.

Newton with distant initial guess (cos(x)-x, x0=10):

The oscillatory cosine sends iterates on an erratic path before entering the basin of attraction, resulting in 92 iterations instead of 3-4.

Fixed-Point g(x) = x^3 - 2:

The fixed-point theorem requires |g'(x*)| < 1. Here g'(x) = 3x^2, so at x*~1.52 we get |g'(x*)| ~ 6.94 >> 1. The iteration is repelling, leading to divergence.

4.3 Theoretical vs. observed convergence rates

Method	Theoretical Rate	Observed
Bisection	Linear, halved/step	Confirmed
FP (|g'|<1)	Linear, ratio |g'(x*)|	0.16 and 0.67
Newton (simple)	Quadratic	2-4 iters
Newton (mult.)	Linear, (m-1)/m	ratio=2/3, m=3

The experimental results match theoretical predictions remarkably well, validating both the implementations and the convergence theory.

5. Conclusion

This assignment demonstrated the trade-offs between reliability and speed in root-finding. Bisection is the safest but slowest. Fixed-Point Iteration depends critically on the choice of g(x). Newton's Method is fastest for simple roots but requires a good initial guess and degrades at multiple roots.

Understanding these behaviors is essential for selecting the right method in practice.

Source files: solvers.py, main.py
Plots: plot_problem1-3.png, plot_fp_divergent.png, plot_newton_ratio_p3.png